

Course Code : ECT 359-3

KQLR/RS – 24 / 1178

**Sixth Semester B. Tech. (Electronics and Communication Engineering)
Examination**

BIOMEDICAL ELECTRONICS

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions are compulsory.
 - (2) All questions carry marks as indicated against them.
 - (3) Assume suitable data and illustrate answers with neat sketches wherever necessary.
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1. (a) What is Man Instrument System ? Describe its various components. Comment on the implications ; If any. 6(CO1)

 (b) Draw and explain the cardiovascular circulation piping diagram. 6(CO1)

 2. (a) Discuss the operation of bio transducers. Classify the bio electrodes used for the measurement of bio potential with special emphasis on their areas of application. 6(CO1,2)

 (b) What are the three kinds of leads from the Einthoven triangle ? Explain. 6(CO2,3)

 3. (a) Explain with a block diagram of a typical set-up for EMG recording. 6(CO2,4)

 (b) Enlist the merits and demerits of direct and indirect blood pressure measuring system. 6(CO1,4)

KQLR/RS-24 / 1178

Contd.

4. (a) Discuss CAT. Write the advantages of CAT over X-ray. 6(CO3)
- (b) Explain the role and responsibilities of a biomedical engineer in an ICU. 6(CO4,5)
5. (a) With reference to shock hazards from electrical equipments, explain electric power distribution system. Support your answer with relevant schematic diagram. 6(CO5)
- (b) Explain the applications of telemetry in patient care. 6(CO4)

